

Project Design Phase

Problem - Solution Fit Template

Date	29 September 2026
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	2 Marks

Problem - Solution Fit:

Problem-Solution Fit checks whether a defined customer problem is meaningfully addressed by the proposed solution. For this project, the fit is assessed against the real passenger journey, operational constraints, available alternatives, user behaviour and the actual ServiceNow implementation.

Purpose

- Understand the current metro ticketing problem from passenger and operations perspectives.
- Identify behavioural triggers, constraints and existing alternatives before validating the proposed solution.
- Check whether the ServiceNow implementation reduces the most important pain points rather than only adding technology.
- Keep implemented functionality separate from future scope so the project remains realistic and verifiable.

Purpose / Vision

Digitize metro ticket booking and validation through ServiceNow so passengers can book faster, receive accurate fares and obtain a verifiable QR-based digital ticket, while metro operations gain centralized, traceable and maintainable workflow automation.

Filled Problem-Solution Fit Analysis

No.	Canvas Section	Project-Specific Findings
1	Customer Segment(s)	• Primary: Metro passengers / daily commuters using self-service digital booking. • Secondary: Station managers and metro operations staff who validate and monitor tickets. • Supporting user: ServiceNow administrators responsible for workflow, roles, ACLs and maintenance.
2	Jobs-to-be-Done / Problems	• Book a metro ticket without depending completely on a physical counter. • Select source, destination, passenger type and ticket quantity correctly. • Know the correct fare before completing the booking. • Receive reliable proof of booking that can be verified at the station. • Track the submitted request and retain booking information digitally. • For operations: reduce manual processing, data mismatch and repeated verification work.
3	Triggers	• Need to travel by metro immediately or during peak hours. • Long queues or slow manual ticketing create a reason to choose self-service. • Passenger wants a paperless ticket available on a phone. • Station/operations teams need faster access to booking and validation data.
4	Emotions: Before / After	• Before: rushed, uncertain about waiting time, concerned about fare accuracy, dependent on counter availability. • During: expects a simple form, clear fare and visible confirmation. • After successful booking: confident, in control, reassured by request status and QR proof. • After successful validation: relieved that the ticket is accepted quickly without paper handling.
5	Available Solutions	• Physical ticket counter: familiar but involves queues, manual handling and paper usage. • Ticket vending / AFC systems: faster than counters but still requires station-side infrastructure. • Existing mobile/web ticketing approaches: convenient but may exist outside the organization's ServiceNow workflow. • Manual records or separate systems: workable for operations but create fragmented data and weaker traceability.
6	Customer Constraints	• Passenger may have limited time during peak travel. • Digital booking should remain simple for users with limited technical knowledge. • Portal behaviour and QR rendering must work reliably within ServiceNow constraints. • Fare logic and station mapping must be accurate to maintain trust. • The first version does not yet include an integrated payment gateway, SMS/WhatsApp ticketing or analytics dashboard.
7	Behaviour	• Passenger opens the ServiceNow portal and selects the metro booking catalog item. • Enters journey details and reviews the calculated fare. • Submits the request and checks status / ticket result. • Opens the QR digital ticket and presents it to station staff. • Operations users verify data, review booking records and use Flow Designer execution details when troubleshooting.
8	Channels of Behaviour	• Online: ServiceNow Service Portal, catalog item, request status, QR widget / spModal, future email/SMS channels. • Offline: metro station, ticket-validation point, interaction with station staff, physical travel environment.
9	Problem Root Cause	• Traditional ticketing depends heavily on manual counters, paper and separate operational steps. • Fare, booking data, workflow and validation are not always handled as one connected digital process. • Manual intervention increases waiting time and creates opportunities for fare or data inconsistencies. • Digital portal environments introduce additional integration challenges such as variable mapping and QR rendering compatibility.

10	Your Solution	• A ServiceNow-based Metro Ticket Generating System using Service Catalog for passenger booking. • Dynamic fare processing based on source, destination and ticket details. • Flow Designer for end-to-end automation and record creation. • Custom Metro Database table for structured booking storage. • QR-based digital ticket rendered through a Service Portal-compatible widget / spModal approach. • Request tracking, ACL / role-based access and Flow Execution Details for operations and troubleshooting. • Future scope: payment gateway, email/SMS/WhatsApp delivery and travel analytics.
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Problem-Solution Fit Evaluation

Fit Dimension	Assessment	Evidence from Project
Customer-Problem Fit	Strong	Passengers need faster booking, correct fare and reliable digital proof; operations need centralized records and lower manual effort.
Problem-Behaviour Fit	Strong	The digital flow matches a natural commuter sequence: select journey -> review fare -> submit -> receive QR -> present for validation.
Problem-Constraint Fit	Good	The implementation is contained within ServiceNow and uses low-code components, but portal QR compatibility and accurate mappings require careful configuration.
Solution-Problem Fit	Strong	Service Catalog, fare logic, Flow Designer, custom Metro Database, QR ticket and request tracking directly address the defined pain points.
Scope Fit	Controlled	Payment gateway, SMS/email/WhatsApp ticketing and analytics are intentionally left as future enhancements rather than overstating the current build.

Final Problem-Solution Fit Statement

The Metro Ticket Generating System demonstrates a clear problem-solution fit because the implemented ServiceNow workflow directly responds to the main passenger and operational problems identified during ideation: queue dependence, manual processing, fare uncertainty, fragmented booking data and the need for verifiable digital tickets. The strongest fit comes from combining a self-service catalog experience with automated fare processing, Flow Designer, structured ticket storage and QR-based validation. Current limitations are treated as future scope rather than hidden inside the implemented feature set.

Current Limitations / Future Validation

- Payment gateway integration is not part of the current implemented core.
- SMS, email and WhatsApp ticket delivery remain future channels.
- Travel analytics and passenger behaviour dashboards remain future extensions.
- Further user testing can validate ease of use, QR reliability and fare accuracy across more routes and passenger scenarios.